

① Selection sort Question

input :-

[18, 22, 20, 19, 21]

int n = 5

i = 0 → 0 < 4

└ L minindex = 0;

└ j = 1 → 1 < 5

└ 22 < 18 → ✗

~~temp = arr[0]~~
arr[0] = arr[0]

└ j = 2 → 2 < 5

└ 20 < 18 → ✗

└ j = 3 → 3 < 5

└ 19 < 18 → ✗

└ j = 4 → 4 < 5

└ 21 < 18 → ✗

∴ minindex = 0

int temp = arr[0] → temp = 18

arr[0] = arr[0] → arr[0] = 18

arr[0] = temp → arr[0] = 18

→ [18, 22, 20, 19, 21]

i = 1 → 1 < 4 → minindex = 1

└ j = 2 → 2 < 5

└ 20 < 22 → ✓ minindex = 2

└ j = 3 → 3 < 5

└ 19 < 20 → ✓ minindex = 3

└ j = 4 → 4 < 5

└ 21 < 19 → ✗

minindex = 3

temp = arr[1] → temp = 22

arr[1] = arr[3] → arr[1] = 19

arr[minindex] = temp → arr[3] = 22

[18, 19, 20, 22, 21]

i=2

→ 2 < 4 → minindex = 2

→ j=3 → 3 < 5

→ arr[3] < arr[2] → 22 < 20 → X

→ j=4 → 4 < 5

→ arr[4] < arr[2] → 21 < 20 → X

∴ minindex = 2

∴ temp = arr[2]

→ temp = 20

arr[2] = arr[minindex] → arr[2] = 20

arr[minindex] = temp → arr[minindex] = 20

[18, 19, 20, 22, 21]

i=3 → 3 < 4 → minindex = 3

→ j=4 → 4 < 5

→ arr[4] < arr[3] → 21 < 22 → ✓

minindex = 4

minindex = 4

∴ temp = arr[3] → temp = 22

arr[3] = arr[4] → arr[3] = 21

arr[4] = temp → arr[4] = 22

[18, 19, 20, 21, 22]

output: - [18, 19, 20, 21, 22]

② Input :- [34, 12, 56, 45, 23]

int n = 5

int i = 0

i = 0 $\rightarrow 0 < n-1 \rightarrow 0 < 4 \rightarrow \text{minindex} = 0$

$\rightarrow j = 1 \rightarrow 1 < 5$

$\hookrightarrow \text{arr}[1] < \text{arr}[0] \rightarrow 12 < 34 \rightarrow \text{minindex} = 1$

$\rightarrow j = 2 \rightarrow 2 < 5$

$\hookrightarrow \text{arr}[2] < \text{arr}[1] \rightarrow 56 < 12 \rightarrow \times$

$\rightarrow j = 3 \rightarrow 3 < 5$

$\hookrightarrow \text{arr}[3] < \text{arr}[1] \rightarrow 45 < 12 \rightarrow \times$

$\rightarrow j = 4 \rightarrow 4 < 5$

$\hookrightarrow \text{arr}[4] < \text{arr}[1] \rightarrow 23 < 12 \rightarrow \times$

$\therefore \text{minindex} = 1$

temp = arr[0] $\rightarrow$ temp = 34

arr[0] = arr[1] arr[0] = 12

arr[1] = temp arr[1] = 34

[12, 34, 56, 45, 23]

i = 1 $\rightarrow 1 < 4 \rightarrow \text{minindex} = 1$

$\rightarrow j = 2 \rightarrow 2 < 5$

$\hookrightarrow \text{arr}[2] < \text{arr}[1] \rightarrow 56 < 34 \rightarrow \times$

$\rightarrow j = 3 \rightarrow 3 < 5$

$\hookrightarrow \text{arr}[3] < \text{arr}[1] \rightarrow 45 < 34 \rightarrow \times$

$\rightarrow j = 4 \rightarrow 4 < 5$

$\hookrightarrow \text{arr}[4] < \text{arr}[1] \rightarrow 23 < 34 \rightarrow \checkmark$

minindex = 4

minindex = 4

temp = arr[1] $\rightarrow$ temp = 34

arr[1] = arr[4] arr[1] = 23

arr[4] = temp arr[4] = 34

[12, 23, 56, 45, 34]

$i=2 \rightarrow 2 < 4 \rightarrow \text{minindex} = 2$

$j=1 \rightarrow 3 < 5$

$\hookrightarrow \text{arr}[1] < \text{arr}[2] \rightarrow 4 < 5 \rightarrow \text{minindex} = 3$

$j=4 \rightarrow 4 < 5$

$\hookrightarrow \text{arr}[4] < \text{arr}[3] \rightarrow 3 < 4 \rightarrow \text{minindex} = 4$

$\text{minindex} = 4$

$\text{temp} = \text{arr}[0] \rightarrow \text{temp} = 56$

$\text{arr}[0] = \text{arr}[i] \quad \text{arr}[i] = 35$

$\text{arr}[i] = \text{temp} \quad \text{arr}[5] = 56$

$[12, 23, 35, 45, 56]$

$i=3 \rightarrow 3 < 4 \rightarrow \text{minindex} = 3$

$j=4 \rightarrow 4 < 5$

$\hookrightarrow \text{arr}[4] < \text{arr}[3] \rightarrow 5 < 4 \rightarrow \text{X}$

$\text{minindex} = 3$

$\text{temp} = \text{arr}[3] \rightarrow \text{temp} = 35$

$\text{arr}[i] = \text{arr}[3] \quad \text{arr}[3] = 45$

$\text{arr}[3] = \text{temp} \quad \text{arr}[1] = 35$

$[12, 23, 35, 45, 56]$

②

Bubble sort

input :- [30, 25, 27, 35, 29]

int i = 0 → 0 < 4

swapped = false;

↳ j = 0 → 0 < 5 - 0 - 1 → 0 < 4

↳ if (arr[0] > arr[1]) { → 30 > 25 ✓

temp = arr[0] → temp = 30

arr[0] = arr[1] arr[0] = 25

arr[1] = temp arr[1] = 30

swapped = true;

}

∴ ~~[25, 30, 27, 35, 29]~~ → [25, 30, 27, 35, 29]

j = 1 → 1 < 4

↳ if (arr[1] > arr[2]) { → 30 > 27 ✓

temp = arr[1] → temp = 30

arr[1] = arr[2] arr[1] = 27

arr[2] = temp arr[2] = 30

swapped = true;

}

→ [25, 27, 30, 35, 29]

j = 2 → 2 < 4

↳ if (arr[2] > arr[3]) { 30 > 35 ✗

temp = arr[2] → temp = 30

arr[2] = arr[3]

arr[3] = temp

}

[25, 27, 30, 35, 29]

$j=3 \rightarrow 3 < 4$
 $\hookrightarrow \text{if}(\text{arr}[3] > \text{arr}[4]) \{ \rightarrow 15 > 29 \checkmark$
 $\quad \text{temp} = \text{arr}[3] \quad \rightarrow \text{temp} = 15$
 $\quad \text{arr}[3] = \text{arr}[4] \quad \text{arr}[3] = 29$
 $\quad \text{arr}[4] = \text{temp} \quad \text{arr}[4] = 15$

$\{$
 $\therefore [25, 27, 10, 29, 15]$
 $\}$

$i=1 \rightarrow 1 < 4$
 $\text{boolean swap} = \text{false};$
 $\hookrightarrow j=0 \rightarrow 0 < 3$
 $\hookrightarrow \text{if}(\text{arr}[0] > \text{arr}[1]) \{ \rightarrow 25 > 27 \quad \times$

$\{$
 $j=1 \rightarrow 1 < 3$
 $\hookrightarrow \text{if}(\text{arr}[1] > \text{arr}[2]) \{ \rightarrow 27 > 30 \quad \times$

$\{$
 $j=2 \rightarrow 2 < 3$
 $\hookrightarrow \text{if}(\text{arr}[2] > \text{arr}[3]) \{ \rightarrow 30 > 29 \checkmark$
 $\quad \text{temp} = \text{arr}[2] \quad \text{temp} = 30$
 $\quad \text{arr}[2] = \text{arr}[3] \quad \text{arr}[2] = 29$
 $\quad \text{arr}[3] = \text{temp} \quad \text{arr}[3] = 30$
 $\quad \text{swapped} = \text{true};$

$\{$
 $\therefore [25, 27, 29, 30, 15]$
 $\}$

$i=2 \rightarrow 2 < 4$
 $\text{boolean swap} = \text{false};$
 $\hookrightarrow j=0 \rightarrow 0 < 2$
 $\hookrightarrow \text{if}(\text{arr}[0] > \text{arr}[1]) \{ 25 > 27 \rightarrow \times$

$\{$
 $j=1 \rightarrow 1 < 2$
 $\hookrightarrow \text{if}(\text{arr}[1] > \text{arr}[2]) \{ 27 > 30 \rightarrow \times$

$\{$
 $\text{swap} = \text{false} \rightarrow \text{break};$
 $\text{break};$

$[25, 27, 29, 30, 15]$

② input 2 = [15, 10, 25, 40, 30]

int i = 0 → 0 < 4

↳ swapped = false

↳ j = 0 → 0 < 4

↳ if (arr[0] > arr[1]) { → 15 > 10 → ✓

temp = 15

arr[0] = 10

arr[1] = 15

swapped = true;

}

[10, 15, 25, 40, 30]

j = 1 → 1 < 4

↳ if (arr[1] > arr[2]) { → 15 > 25 ✗

}

j = 2 → 2 < 4

↳ if (arr[2] > arr[3]) { → 25 > 40 ✗

}

j = 3 → 3 < 4

↳ if (arr[3] > arr[4]) { → 40 > 30 ✓

temp = 40

arr[3] = 30

arr[4] = 40

}

[10, 15, 25, 30, 40]

i = 1 → 1 < 4

↳ swapped = false

↳ j = 0 → 0 < 3

↳ if (arr[0] > arr[1]) { 10 > 15 → ✗

↳ j = 1 → 1 < 3

↳ if (arr[1] > arr[2]) { 15 > 25 → ✗

↳ j = 2 → 2 < 3

↳ if (arr[2] > arr[3]) { 25 > 30 → ✗


```
swapped = false;
break;
```

→

[10, 15, 25, 30, 50]

③ Insertion Sort

① input :- [18, 22, 20, 19, 21]

ini i = 1 → 1 < 5

currentElem = 22

left = 0

18 > 22

while (left > 0 & arr[left] > currentElem) { X

}

arr[1] = 22

}

→ [18, 22, 20, 19, 21]

i = 2 → 2 < 5

currentElem = 20

left = 1

while (left > 0 & arr[left] > 20) { ✓

arr[2] = arr[1]

left = 0

→ [18, 22, 22, 19, 21]

}

arr[1] = ~~22~~ 20 → [18, 20, 22, 19, 21]

}

→ [18, 20, 22, 19, 21]

~~~~~



j=3

3 < 5

currentEle = 19

left = 2

while (27 < 0 44 22 7 19) { ✓

arr[3] = arr[2] → [18, 20, 22, 22, 21]

left = 1

}

while (17 < 0 44 20 7 19) {

arr[2] = arr[1] → [18, 20, 20, 22, 21]

left = 0

}

while (07 < 0 44 10 7 19) { ✗

}

∴ arr[1] = 19 → [18, 19, 20, 22, 21]

i=4 → 4 < 5

currentEle = 22

left = 3

while (37 < 0 44 22 7 21) { ✓

arr[4] = arr[3] → [18, 19, 20, 22, 22]

left = 2

}

while (27 < 0 44 20 7 21) { ✗

}

arr[3] = 21 → [18, 19, 20, 21, 22]

output: [18, 19, 20, 21, 22]



② input = [38, 12, 56, 45, 23]

int i = 0 → 4 < 5

currentEle = 12

left = 0

while (0 > 0 44 arr[0] > 12) { → ✓

arr[1] = 38

→

[38, 38, 56, 45, 23]

left = -1

}

arr[0] = 12 → [12, 38, 56, 45, 23]

while (-1 > 0 44 arr[-1] > 12) ✗

i = 2 → 2 < 5

currentEle = 56

left = 1

while (1 > 0 44 38 > 56) ✗

}

arr[2] = 56 → [12, 38, 56, 45, 23]

i = 3 → 3 < 5

currentEle = 45

left = 2

while (2 > 0 44 56 > 45) ✓

arr[3] = arr[2]

→

[12, 38, 56, 56, 23]

left = 1

}

while (1 > 0 44 38 > 45) ✗

}

arr[2] = 45 → [12, 38, 45, 56, 23]



$i=4 \rightarrow 4 < 5$

currentElem = 23

left = 3

while (3 > 0 44 56 723) { ✓

arr[4] = arr[3] → [12, 38, 45, 56, 56]

left = 2;

}

while (2 > 0 44 45 723) { ✓

arr[3] = arr[2] → [12, 38, 45, 45, 56]

left = 1

}

while (1 > 0 44 38 723) {

arr[2] = arr[1] → [12, 38, 38, 45, 56]

left = 0

while (0 > 0 44 12 723) { ✗

}

arr[1] = 23

→ [12, 23, 38, 45, 56]

output:- [12, 23, 38, 45, 56]



# ④ Merge Sort

① input = [50, 23, 9, 18, 61, 32]

```
public static void f(arr, 0, 5) {
```

```
    if (0 > 5) { return; } X
```

```
    mid = 2
```

```
    f(arr, 0, 2) → [50, 23, 9] [9, 23, 50]
```

```
    f(arr, 3, 5) → [18, 32, 61]
```

```
    merge(arr, low, mid, high) →
```

```
    merge(arr, 0, 2, 5) → [9, 18, 23, 32, 50, 61]
```

```
}
```

```
→ f(arr, 0, 2) {
```

```
    if (0 > 2) X
```

```
    mid = 1
```

```
    f(arr, 0, 1) → [23, 50]
```

```
    f(arr, 1, 2) → return; → [9, 23]
```

```
    merge(arr, 0, 1, 2) → [9, 23, 50]
```

```
}
```

```
→ f(arr, 0, 1) {
```

```
    if (0 > 1) X
```

```
    mid = 0
```

```
    f(arr, 0, 0) → return
```

```
    f(arr, 1, 1) → return;
```

```
    merge(arr, 0, 0, 1) → [23, 50]
```

```
}
```

```
→ f(arr, 3, 5) {
```

```
    if (3 > 5) X
```

```
    mid = 4
```

```
    f(arr, 3, 4) → [18, 61]
```

```
    f(arr, 5, 5) → return
```

```
    merge(arr, 3, 4, 5) → [18, 32, 61]
```

```
}
```

```
→ f(arr, 3, 5) {
```

```
    mid = 3
```

```
    f(arr, 3, 3) → return
```

```
    f(arr, 4, 4) → return
```

```
} - [18, 61]
```



[9, 23, 50, 13, 12, 11]

merge(arr, 0, 2, 5) {

blue = 0

green = 3, red = 0

while (0 <= 2 && 3 <= 5) { ✓

if (9 < 18)

merge[0] = 9 → [9]

red = 1

blue = 1

while (1 <= 2 && 3 <= 5) {

if (23 < 18) X

merge[1] = 13 → red = 2, blue = 4 → [9, 13]

while (1 <= 2 && 4 <= 5) {

if (23 < 12) X

merge[2] = 23 → red = 1, green = 4 → [9, 13, 23]

while (2 <= 2 && 4 <= 5) {

if (50 < 12) X

merge[3] = 50 → [9, 13, 23, 50, 12]

merge(arr, 0, 1, 2, 1) { → [50, 23, 9]

blue = 0

green = 2

red = 0

while (0 <= 1 && 2 <= 2) { ✓

if (50 <= 9) X

merge[0] = arr[2]

red = 1

green = 3

→ [9] ~~23, 50~~

while (0 <= 1 && 3 <= 2) { X

while (0 <= 1) {

arr[1] = arr[0] → [9, 23, 50]

}

merge(arr, 3, 4, 5) {

blue = 3

green = 4

red = 0

while (3 <= 4 && 5 <= 5) { ✓

if (13 <= 32) X

merge[red] = arr[4] → [13, 61, 32]

red = 1

blue = 4

while (4 <= 4 && 5 <= 5) { ✓

if (61 <= 32) X

merge[2] = arr[5] → [13, 32, 61]

}



② ↑ low ↑ high  
input = [12, 45, 1, 23, 67, 90]

```
public static void f(arr, 0, 5) {
    mid = if (0 > 5) { — X
        mid = 2
```

f(arr, 0, 2) → [1, 12, 45]

f(arr, 3, 5) → [23, 67, 90]

merge(arr, 0, 2, 5) →

}

f(arr, 0, 2) {

if (0 > 2) X

mid = 1

f(arr, 0, 1) → [12, 45]

f(arr, 2, 2) → return

merge(arr, 0, 1, 2) → [1, 12, 45]

}

f(arr, 0, 1) {

if (0 > 1) X

mid = 0

f(arr, 0, 0) → return

f(arr, 1, 1) → return

merge(arr, 0, 0, 1)

}

f(arr, 3, 5) {

if (3 > 5) {

mid = 4

f(arr, 3, 4) → [23, 67]

f(arr, 5, 5) → return

merge(arr, 3, 4, 5) → [23, 67, 90]

}

f(arr, 3, 4) {

mid = 3

f(arr, 3, 3) → return

f(arr, 4, 4) → return

merge(arr, 3, 3, 4) → [23, 67]

}



[1, 12, 45, 23, 67, 90]

merge(arr, 0, 2, 5) {

int merge = new int [5 - 0 + 1] = [6]

blue = 0

green = 3

red = 0

while (0 <= 2 && 3 <= 5) {

if (1 <= 2) {

merge[0] = 1 → [1]

red = 1

blue = 1

}

while (1 <= 2 && 3 <= 5) {

if (12 <= 23) {

merge[1] = 12 → [1, 12]

red = 2

blue = 2

}

while (2 <= 2 && 3 <= 5) {

if (45 <= 23) ✗

merge[2] = 23 → [1, 12, 23]

red = 3

green = 4

}

while (2 <= 2 && 4 <= 5) {

if (45 <= 67) {

merge[3] = 45 → [1, 12, 23, 45]

red = 4

blue = 3

}

while (3 <= 2) ✗

while (4 <= 5) {

merge[4] = 67 → [1, 12, 23, 45, 67]

red = 5

green = 5

}

while (5 <= 5) {

merge[5] = 90 → [1, 12, 23, 45, 67, 90]

red = 6

green = 6

}



for

 $\text{int } i = 0 \rightarrow 0 < 6$  $\text{arr}[0+0] = \text{merge}[0] \rightarrow [1, 45, 1, 23, 67, 90]$  $i = 1 \rightarrow 1 < 6$  $\hookrightarrow \text{arr}[1] = \text{merge}[1] \rightarrow [1, 12, 1, 23, 67, 90]$  $i = 2 \rightarrow 2 < 6$  $\hookrightarrow \text{arr}[2] = \text{merge}[2] \rightarrow [1, 12, 23, 23, 67, 90]$  $i = 3 \rightarrow 3 < 6$  $\hookrightarrow \text{arr}[3] = \text{merge}[3] \rightarrow [1, 12, 23, 45, 67, 90]$  $i = 4 \rightarrow 4 < 6$  $\hookrightarrow \text{arr}[4] = \text{merge}[4] \rightarrow [1, 12, 23, 45, 67, 90]$  $i = 5 \rightarrow 5 < 6$  $\hookrightarrow \text{arr}[5] = \text{merge}[5] \rightarrow [1, 12, 23, 45, 67, 90]$  $\rightarrow \text{output}:- [1, 12, 23, 45, 67, 90]$



# ⑤ Quick sort

① {75, 85, 55, 95, 65}

{55, 65, 75, 85, 95}

[55, 65]

[75, 85]

[55] [65]

[75] [85]

```

psum { arr, 0, 4 } {

```

```

    if (0 == 4) X

```

```

    int pivot = pivotfinder(arr, low, high)

```

```

    f(arr, low, pivot-1)

```

```

    f(arr, pivot+1, high)

```

```

int pivotfinder(arr, 0, 4) {

```

```

    int pivot = arr[0]

```

```

    i = 0

```

```

    j = 4

```

```

    while (i < j) {

```

```

        while (i <= high && arr[i] <= pivot) {

```

```

            i++

```

```

        }

```

```

        while (j >= low && arr[j] > pivot) {

```

```

            j--

```

```

        }

```

```

        if (i < j) {

```

```

            temp = arr[i]

```

```

            arr[i] = arr[j]

```

```

            arr[j] = temp

```

```

        return i

```

```

    }

```



[75, 85, 55, 95, 65]

F(arr, 0, 4)

↳ int pivot = pivotFindiv (0, 4)

↳ pivotFindiv (0, 4)

    pivot = 75

    int i = 0

    j = 4

    while (i < j) {

        while (i < j & arr[i] <= 75) {

            i++;

        }

        while (j > 0 & arr[j] > pivot) {

            j--;

        }

        if (i < j)

            int temp = arr[i];

            arr[i] = arr[j];

            arr[j] = temp;

    } j = i, i = 1

    { 95, 85, 55, 75, 65 }

    while (i < j) {

        while (i < j & arr[i] <= pivot) {

            i++;

        while (j > 0 & arr[j] > 75) {

            j--;

        }

        temp = arr[i];

        arr[i] = arr[j];

        arr[j] = temp;

→ [55, 85, 95, 75, 65]

j = 2, i = 1

{ 55, 85, 95, 75, 65 }

while (i < j) {

    while (i < j & arr[i] <= 75) {

        i++;

    while (j > 0 & arr[j] > 75) {

        j--;

    }



[75, 85, 55, 95, 65]

F(arr, 0, 4)

↳ int pivot = pivotfindiv(0, 4)

↳ pivotfindiv(0, 4)

    pivot = 75

    int i = 0

    j = 4

    while(i < j) {

        while(i < j & 75 <= arr[j]) {

            i++;

        }

        while(j > 0 & 75 > arr[j]) {

            j--;

        }

        if(i < j)

            int temp = arr[i]

            arr[i] = arr[j]

            arr[j] = temp

    j = 3, i = 1

    {95, 85, 55, 75, 65}

    while(i < j) {

        while(i < j & 75 <= arr[j]) {

            i++;

        while(j > 0 & 75 > arr[j]) {

            j--;

        }

        temp = arr[i]

        arr[i] = arr[j]

        arr[j] = temp

→ [55, 85, 95, 75, 65]

j = 2, i = 1

{55, 85, 95, 75, 65}

while(i < j) {

    while(i < j & 75 <= arr[j]) {

        i++;

    while(j > 0 & 75 > arr[j]) {

        j--;

    }



i = 2, j = 2

if (i < j) {

    pivot = 2

while (i < j) {

    {

        pivot = j    j = 2

        { 55, 65, 75, 85, 95 }

          ↑

          ↑

F(0, 1)

    { int pivot = pivotfinder(0, 1);

        pivot = 55

        int i = low = 0

        j = 1

        while (i < j) {

            while (i <= j && arr[i] <= 55) {

                i++;

            }

            65

55

            while (j >= 0 && arr[j] > pivot) {

                j--;

            }

        i = 1, j = 0

        if (i < j) {

            {

                return j = 0;    pivot = 0;

            { 55, 65, 75, 85, 95 }



② {42, 56, 12, 67, 34}

F(0, 4)

int pivot = pivotFindiv(0, 4)

int pivotFindiv(<sup>0</sup>low, <sup>4</sup>high, arr) {

    pivot = 42

    i = 0

    j = 4

    while(i < j) {

        while(i <= 4 & 44 42 <= pivot) {

            i++;

        }

        i = 2      while(j >= 0 & 44 56 > 42) {

            j--;

        }

    i = 1

    j = 3

    if(i < j) {

        tmp = 42

        arr[i] = 67

        arr[j] = 42

    i = 1, j = 3

    {67, 56, 12, 42, 34}

    while(i < j) {

        while(i <= 4 & 44 <sup>56</sup>arr[i] <= 42) {

            i++;

        }

        while(j >= 0 & <sup>56</sup>44 <sup>32</sup>arr[j] > 42) {

            j--;

        }

    i = 1

    j = 2

    i = 2

    if(i < j) {

        tmp = 56

        arr[i] = 12

        arr[j] = 56



{ 67, 12, 56, 42, 343 }

i=1, j=2

while (1<2) {

while (1<= 4 & 12<= 42) {

i++;

i=2, j=2

while (2770 & 4 & 567, 42) {

j--;

}

i=2, j=1

if (2<1) {

} X

{ 67, 12, 56, 42, 343 }

while (2<3) {

42

while (2<= 4 & arr[i]<=42) {

i++;

}

i=3, j=0

while (37=0 & 4 & 56 > 42) {

j--;

}

{ 12, 34, 42, 56, 17 }



{ 67, 12, 56, 42, 34 }

i=1, j=2

while (1 < 2) {

while (1 <= 4 && 12 <= 42) {

i++;

i=2, j=2

while (2 > 0 && 56 > 42) {

j--;

}

i=2, j=1

if (2 < 1) {

} X

{ 67, 12, 56, 42, 34 }

while (2 < 3) {

42

while (2 <= 5 && arr[i] <= 42) {

i++;

}

i=3, j=0

while (3 > 0 && 56 > 42) {

j--;

}

{ 12, 34, 42, 56, 67 }